Reading the Storm: Media Attribution of United States Climate Disasters, 2000–2024

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June 22, 2026

Abstract

We construct an Attribution Index measuring how U.S. newspaper coverage frames the causal origins of billion-dollar climate disasters. Using an LLM-based coding pipeline validated against an independent human audit ($\kappa = 0.96$ for relevance, weighted $\kappa_w = 0.96$–0.98 for attribution), we classify 1,252 articles retrieved for all 313 NOAA billion-dollar disaster events from 2000 to 2024 on a scale from explicit non-climate attribution (-1) to explicit anthropogenic attribution ($+1$) and aggregate these to a disaster-level index. We document three findings. First, the dominant mode of disaster coverage is silence on causation: 74.8% of articles contain no causal claim, and the mean disaster-level index is 0.028, close to neutral. Second, attribution framing has risen steadily at 0.013 index points per year ($p < 0.001$), shifting from a pre-2010 mean of -0.105 to a post-2010 mean of $+0.057$, consistent with a structural break in partisan climate discourse documented in the prior literature. Third, outlet political lean predicts attribution framing: a one-point increase in Ad Fontes Bias score is associated with a 0.011-point decrease in the Attribution Index ($p = 0.013$, clustered standard errors), an effect concentrated in national and wire-service coverage and robust to disaster fixed effects and alternative lean measures. By contrast, disaster type, CPI-adjusted cost, death toll, and geography have no significant predictive power. These findings establish a new empirical baseline for the role of media framing in the transmission of climate-disaster risk.

Keywords: climate change attribution, media framing, natural language processing, disaster economics

1 Introduction

Between 2000 and 2024, the United States experienced 313 weather and climate disasters each causing at least one billion dollars in CPI-adjusted losses, together inflicting more than $2.36 trillion in damages and 10,849 fatalities. The pace has been accelerating: Boustan et al. (2020) show, using a century of U.S. county data, that the frequency of severe natural disasters has increased significantly since the 1990s, a trend visible in the NOAA record itself. These events are not only physical shocks; they are also information events. When a hurricane makes landfall or a drought devastates a region's agriculture, the resulting newspaper coverage is among the primary channels through which millions of Americans form beliefs about what happened, why, and what it portends.

Whether that coverage attributes an extreme event to anthropogenic climate change or frames it as natural variability is a consequential editorial choice. The economics literature has established that disaster media coverage materially shapes downstream behavior: using flood events as a natural experiment, Gallagher (2014) finds that insurance take-up spikes by approximately 9% in directly affected areas following a flood, with non-flooded communities in the same television media market experiencing a take-up increase roughly one-third as large—evidence that the information channel, not just direct experience, drives risk updating. On the policy side, Magontier (2020) finds that a one-standard-deviation increase in storm media coverage is associated with 54% more FEMA hazard mitigation projects in the affected area, consistent with coverage functioning as a risk signal to governments as well as households. If the *content* of coverage matters—if framing a disaster as a symptom of climate change rather than as a one-off natural event shapes how readers and policymakers update their beliefs about future risk— then the question of what U.S. newspapers actually say about causation is one with real stakes for insurance markets, preparedness investment, and public understanding of climate risk.

Yet we know surprisingly little about the systematic pattern of causal attribution framing across U.S. disaster coverage. Prior work has examined how coverage of climate change broadly has polarized over time—Chinn et al. (2020) show that partisan divergence in climate discourse language was flat through most of the 2000s before a discrete structural break around 2010–2011—and community-level studies have documented that attribution discussion is minimal even after high-impact events (Boudet et al., 2020). But no prior study has produced a systematic, quantitative measure of how causal attribution framing varies across the full universe of major U.S. disaster events, across outlet types, and over time. The barriers are practical: hand-coding attribution framing for hundreds of events and thousands of articles is prohibitively costly.

This paper removes that barrier. We construct an Attribution Index covering 1,252 newspaper articles retrieved for all 313 NOAA billion-dollar disaster events from 2000 to 2024, using a validated large language model (LLM) coding pipeline that follows the three-step text-as-data framework of Gentzkow et al. (2019). Each article is first screened for relevance and then assigned to one of five attribution categories: explicit anthropogenic framing (+1), hedged or contextual climate mention (+0.5), no causal attribution (0), explicit non-climate attribution (−1), or mixed/contested framing (0, separately flagged). The pipeline is validated against an independent human post-hoc audit of 100 articles—blind to the LLM's codes— which achieves $\kappa = 0.96$ for relevance screening and weighted $\kappa_w = 0.96$–0.98 for attribution coding, comfortably exceeding the reliability targets set in advance. Article-level scores are averaged to a disaster-level Attribution Index, enabling systematic regression analysis of the determinants of attribution framing at both the disaster and article level.

We document three main findings. First, the dominant mode of disaster coverage is silence on causation: 74.8% of relevant articles contain no causal attribution at all, and the mean disaster-level Attribution Index is only 0.028—near-neutral, and far from the confident anthropogenic framing (+1) that attribution science might lead one to expect. Second, the level of attribution framing has risen steadily over the sample period at approximately 0.013 index points per year ($p < 0.001$), shifting from a pre-2010 mean of −0.105 (net non-climate framing) to a post-2010 mean of +0.057 (modest net anthropogenic framing). This timing aligns with the structural break in partisan climate discourse identified by Chinn et al. (2020). Third, outlet political lean is a robust predictor of attribution framing: articles from more right-leaning outlets (by Ad Fontes Bias score) receive lower Attribution Index values ($\hat{\beta} = -0.011$ per Bias-score point, $p = 0.013$, clustered standard errors), a result that survives disaster fixed effects, alternative lean measures, and an ordered-logit functional-form check, and is concentrated in national and wire-service coverage. By contrast, none of the disaster's own physical characteristics—type, CPI-adjusted cost, death toll, duration, or geography—predict attribution framing.

The paper proceeds as follows. Section 3 describes the NOAA disaster dataset, the newspaper collection pipeline, and the construction of the Attribution Index. Section 4 details the LLM coding procedure, validation design, and regression specifications. Section 5 presents the main results, and Section 6 reports the robustness checks. Sections 6 and 7 discuss implications and conclude.

2 Related Literature

This paper connects four strands of research: (i) the growing literature on climate communication and media framing of extreme weather; (ii) the economics of natural disasters, salience, and risk perception; (iii) text-as-data and natural language processing methods in economics; and (iv) the political economy of local media markets.

Climate Communication and Media Framing

A central concern in climate communication research is whether and how media coverage links extreme weather events to anthropogenic climate change. Boudet et al. (2020) study community-level responses to 15 high-impact weather events across the United States, analyzing over 4,610 newspaper articles and 164 community interviews. They find that local media discussion of climate change is minimal even in the aftermath of events with a strong scientific attribution signal — fewer than five attribution-related articles per community on average. Crucially, when attribution coverage does appear, both partisanship and event-type confidence shape the discussion. The paper distinguishes three tiers of scientific confidence in event attribution: zero confidence for extratropical cyclones, tornadoes, and mudslides; intermediate confidence for droughts, wildfires, flooding, and tropical cyclones; and high confidence for extreme heat and cold events specifically. Even high-confidence events do not reliably generate sustained attribution discourse at the local level. Our paper extends this agenda from the small-N qualitative approach of Boudet et al. (2020) to a large-N quantitative analysis, trading community-interview depth for systematic coverage of all 313 NOAA billion-dollar disaster events over a 25-year panel.

Chinn et al. (2020) analyze U.S. climate news coverage over 1985–2017 using Wordfish, an unsupervised text-scaling method, to measure the ideological positioning of climate discourse over time. They find that politicization of climate news rose gradually throughout the period, but that partisan polarization in coverage language was relatively flat until a discrete structural break in 2010–2011 — precisely the pre-/post-2010 break visible in our Figure 1 and confirmed by our year-trend regression. Their key outcome is *discourse polarization* — the divergence between outlets in the partisan language used to discuss climate — while ours is *causal attribution framing* — whether coverage attributes a specific event to climate change or to natural variability. The two complement each other: polarization in language is a necessary but not sufficient condition for divergence in causal attribution. Chinn et al. (2020) also document that political actors outnumber scientists as quoted sources in climate coverage from 2006 onward, consistent with the supply-side political-economy mechanism we revisit in the outlet-lean analysis (Section 5.4).

Carmichael et al. (2017) examine the drivers of public concern about climate change using quarterly time-series data over 2001–2014. Contrary to a simple salience hypothesis, they find that extreme weather events do not significantly increase aggregate climate concern. Instead, media attention and political elite cues are the dominant drivers of public opinion dynamics. They document echo chamber and boomerang effects in which partisan media exposure can *decrease* concern among ideologically unsympathetic audiences. Our outlet-lean finding is consistent with this mechanism at the supply side: if right-leaning outlets systematically produce less anthropogenic attribution framing, ideologically sorted audiences are exposed to less such framing, reinforcing the partisan gap in concern identified by Carmichael et al. (2017).

Hai and Perlman (2022) use a preregistered two-wave survey experiment (approximately 9,174 respondents total) to examine how politicians' public attribution of extreme weather to climate change affects public perception. They find a backfire effect among Republicans, who rate attribution-making politicians as *less* capable. At the same time, they document a modest increase over time in the frequency with which politicians make explicit attribution claims. These results imply a political-economy constraint on attribution language: political actors face differential electoral incentives to link extreme events to climate change, and those incentives vary by party and constituency. This mechanism is consistent with the partisan outlet-lean gradient we document at the media level.

Economics of Natural Disasters, Salience, and Policy

Three papers from the economics literature speak directly to how disaster coverage translates into behavioral and policy responses. Gallagher (2014) provides the most direct evidence of a media-salience channel, showing that flood insurance take-up spikes by approximately 9% following a flood event in the affected area, then decays over a nine-year window. Critically, non-flooded communities in the *same* television media market experience a take-up increase one-third as large as directly affected communities — clean evidence of a media-driven information transmission effect that operates through coverage rather than direct experience. Our paper measures the climate-attribution content of exactly the kind of coverage that Gallagher (2014) shows is behaviorally consequential.

Boustan et al. (2020) document the macroeconomic consequences of natural disasters using a century of U.S. county data. They find that severe disasters increase out-migration by 1.5 percentage points and depress housing prices by 2.5–5%; they further show that disaster frequency has accelerated since the 1990s. This acceleration provides the economic backdrop for our sample: the disasters we study are a growing risk, making the question of how media covers their causes increasingly consequential.

Magontier (2020) provides the most direct empirical link between disaster media coverage and government disaster-preparedness behavior. Using variation in storm media coverage across U.S. markets, he finds that a one-standard-deviation increase in storm media coverage is associated with 54% more FEMA hazard mitigation projects in the affected area. He proposes a risk-signaling mechanism in which media coverage serves as a credible signal of disaster risk to voters and policymakers, driving preparedness investment. If the *content* of coverage matters — whether a disaster is framed as a consequence of climate change or as a natural event — then the attribution framing we measure may modulate the character of this policy response; this is a question our paper motivates but does not directly answer.[1]

Egan and Mullin (2012) provide a complementary micro-level result at the individual level. Exploiting within-person variation in local temperature departures from seasonal norms, they find that a 3.1°F increase in local temperature above its seasonal norm raises the probability that a respondent says there is "solid evidence" of warming by approximately 1 percentage point, with the effect decaying within 12 days. The effect is largest among respondents who are less educated and who have weak attachments to political parties. This evidence that direct environmental experience shapes climate beliefs through a short-term salience mechanism complements our evidence that media attribution framing shapes the broader information environment in which those experiences are interpreted.

Text as Data and NLP Methods in Economics

The methodological approach of this paper — using a large language model to classify natural language text at scale — sits within a growing economics literature on text-as-data methods. Gentzkow et al. (2019) provide a comprehensive survey of this literature, organizing it around a three-step framework: (i) represent text as a numerical array (via bag-of-words, embeddings, or model outputs); (ii) map those representations to predicted values or classifications; and (iii) use the predicted values in subsequent economic analysis. Our pipeline maps directly onto this framework, with steps (i)–(ii) implemented by a large language model (Claude, Anthropic) following a written codebook, and step (iii) implemented by the regression analysis in Section 5.

Gentzkow and Shapiro (2010) represent the paradigmatic application of text-as-data methods to measure media bias, using the frequency of partisan phrases in congressional speech to construct a slant index for over 400 U.S. daily newspapers. They find that approximately 20% of the variation in outlet slant is explained by reader demographic demand, while

[1]Magontier (2020) is currently a working paper; its empirical findings should be interpreted with appropriate caution pending peer review.

owner identity explains far less. This result disciplines our interpretation of the outlet-lean gradient we document: partisan framing differences across outlets appear to reflect audience demand as much as proprietorial preferences. Their approach uses the text of newspapers as the object of analysis, exactly as we do, though they measure partisan language use while we measure causal attribution framing.

Political Economy of Local Media Markets

Our finding that the political-lean effect is concentrated in national-outlet coverage raises questions about the role of local media market structure more broadly. Ellger et al. (2024) study the causal effect of local newspaper closures on political polarization using German newspaper exits over 1980–2009, exploiting exit events as quasi-experimental variation in local newspaper supply. Using a difference-in-differences design, they find that local newspaper exits increase political polarization, and that the mechanism operates through substitution: when a local paper exits, consumers shift to national tabloid coverage that is more nationally partisan and less locally focused. This result offers a candidate explanation for the pattern we document — local-outlet attribution framing shows no significant lean gradient (Section 5.4), while national-outlet framing shows a clear partisan pattern — if local papers play a distinct informational role that is not replicable by national substitutes, their ongoing decline could gradually shift the aggregate information environment toward the nationally partisan framing that dominates in our national-outlet subsample.

3 Data

3.1 NOAA Billion-Dollar Disasters Dataset

The sample is drawn from NOAA's Weather and Climate Billion-Dollar Disasters dataset, restricted to U.S. events occurring between 2000 and 2024 (313 events). Each event is characterized by a disaster type (Severe Storm, Tropical Cyclone, Flooding, Drought, Wildfire, Winter Storm, or Freeze), a begin and end date, the set of affected states, CPI-adjusted and unadjusted cost (in millions of 2024 dollars), and a fatality count. Severe storms are the most common event type (179 of 313 events), followed by tropical cyclones (48), flooding (30), drought (21), wildfire (19), winter storm (13), and freeze (3). Total CPI-adjusted damages across the sample exceed \$2.36 trillion, with a median event cost of \$2.4 billion (mean \$7.5 billion, driven by a right tail of extreme events up to \$201.3 billion); total reported fatalities across all events are 10,849.

3.2 Newspaper Coverage Data

Newspaper universe. The candidate universe of outlets is drawn from the Access World News (AWN) database, which lists 3,647 U.S. newspapers and wire services with associated city, state, and language metadata. Of these, 970 had no listed website and were excluded. The remaining outlets were passed through an automated verification pipeline that attempted to resolve and HTTP-check each listed URL (classifying each as verified, verified via HTTP-only, or unverified with a status code/reason); 899 outlets ultimately produced a usable, live domain.

Programmable Search Engine (PSE) construction. The 899 verified domains were partitioned into 18 regional Google Programmable Search Engines (PSEs), grouped by the states each outlet primarily serves, plus a 19th "National" PSE covering major national and wire outlets (e.g., the *New York Times*, *Washington Post*, CNN, AP, NPR, USA Today) that are relevant regardless of event geography. Each of the 313 disaster events is mapped to the subset of regional PSEs corresponding to its affected states (plus the National PSE), so that searches are restricted to outlets plausibly covering that event's geography.

Collection pipeline. For each event, queries combining the disaster name/type and date window are issued against the relevant PSE subset via the Google Custom Search JSON API. Candidate results are filtered on (i) a date window around the event's begin/end dates (with retrospective coverage allowed up to the start of the next event of the same type, where applicable), (ii) geographic relevance – results must reference a place name consistent with the event's affected states or broader region, and (iii) disaster-type/keyword relevance. Up to 30 candidate articles are retained per event. The full collection run returned 1,694 candidate articles across the 313 events: 237 events (75.7%) returned at least one candidate article (mean 5.4 articles per event among events with any coverage; 76 events returned none). Of the 1,694 candidate articles, 911 (53.8%) are from outlets on the National PSE list and 783 (46.2%) are from regional/local outlets. These candidate articles are then passed through the relevance-screening and attribution-coding procedure described in Section 4.

3.3 Constructing the Attribution Index

Each of the 1,694 candidate articles is first screened for *relevance* – whether it substantively discusses the specific NOAA-listed event it was retrieved for (correct disaster, geography, and time window), versus being off-topic, ambiguous (`Borderline`), or unusable (`Trash`). Only articles coded `Relevant` proceed to attribution coding.

Each relevant article is then assigned to one of five categories describing how it frames the causal origin of the event, with an associated numeric score $a_{i,d} \in \{-1, 0, 0.5, 1\}$:

Table 1: Attribution Coding Categories

Code	Description	Score
A	Explicit anthropogenic climate-change attribution	+1.0
B	Hedged / contextual climate mention	+0.5
C	No causal attribution (pure event reporting)	0.0
D	Explicit non-climate attribution (natural variability or other human causes)	−1.0
E	Mixed / contested framing	0.0 (flagged)

Notes: Full category definitions, decision rules, and example language are given in the project codebook (`paper/attribution_codebook.md`). Category E receives the same numeric score as C but is separately flagged as *contested* when computing disaster-level dispersion.

Article-level scores are averaged to a disaster-level *Attribution Index*; the aggregation formula, calibration procedure, and treatment of events with no relevant coverage are described in Section 4.

3.4 Summary Statistics

Table 2 summarizes the collection-stage data described above, together with the results of the relevance-screening and attribution-coding pass described in Section 4. Of the 1,694 candidate articles, full text was successfully retrieved for 1,591 (93.9%); of those, 1,252 (78.7%) were coded `Relevant` to the specific event they were retrieved for, with the remainder split between `Borderline` (210, 13.2%) and `Trash` (129, 8.1%). Among Relevant articles, the large majority (937, 74.8%) contain no causal attribution at all (category C, pure event reporting); explicit anthropogenic attribution (A) and hedged/contextual climate mentions (B) together account for 252 articles (20.1%), explicit non-climate attribution (D) for 53 (4.2%), and mixed/contested framing (E) for only 10 (0.8%) – contested coverage is rare both at the article level and at the disaster level (only 6 of the 209 covered disasters, 2.9%, have any Contested article at all). At the disaster level, 209 of the 313 events (66.8%) have at least one Relevant article (mean 6.0, median 4 Relevant articles per covered event); the remaining 104 (33.2%) have no Relevant coverage at all and are analyzed as a separate no-attribution group rather than dropped (Section 4). The mean disaster-level Attribution Index among covered events is 0.028 (median 0.000, SD 0.241, range $[-1, 1]$) – i.e. coverage is, on average, only very mildly tilted toward anthropogenic framing and is dominated in practice by events with little to no causal framing either way, consistent with the category-C share above.

Table 2: Summary Statistics: Sample, Coverage, and Coding Outcomes

	Value
NOAA billion-dollar disaster events (2000–2024)	313
Candidate newspaper articles collected	1,694
Events with ≥ 1 candidate article	237 (75.7%)
Events with no candidate articles	76 (24.3%)
Mean articles per covered event	5.4
Maximum articles for a single event	30
Articles from National PSE outlets	911 (53.8%)
Articles from regional/local PSE outlets	783 (46.2%)
Full-text retrieval and coding outcomes:	
Candidate articles with full text retrieved ("found")	1,591 (93.9%)
Articles coded `Relevant` (of found)	1,252 (78.7%)
Articles coded `Borderline` (of found)	210 (13.2%)
Articles coded `Trash` (of found)	129 (8.1%)
Category A – explicit anthropogenic (of Relevant)	107 (8.5%)
Category B – hedged/contextual climate mention	145 (11.6%)
Category C – no causal attribution	937 (74.8%)
Category D – explicit non-climate attribution	53 (4.2%)
Category E – mixed/contested	10 (0.8%)
Disaster-level Attribution Index:	
Disasters with ≥ 1 Relevant article ($N_d > 0$)	209 (66.8%)
Disasters with no Relevant coverage ($N_d = 0$)	104 (33.2%)
Mean / median N_d among covered disasters	6.0 / 4
Mean AttributionIndex$_d$ (covered disasters)	0.028
Median / SD AttributionIndex$_d$	0.000 / 0.241
Disasters with any Contested coverage	6 (2.9% of covered)

Notes: Coverage statistics in the top panel describe the raw candidate-article sample prior to relevance screening. The bottom two panels reflect the full-sample LLM coding pass described in Section 4; reliability statistics from the independent post-hoc audit sample are reported separately once available (Section 4).

4 Empirical Approach

4.1 Article-Level Classification

Each candidate article is coded in two sequential steps, following a written codebook (reproduced in full in the project repository as `attribution_codebook.md`) that is also used as the basis for the LLM coding prompt.

Step 1: Relevance screening. Every candidate article is classified as `Relevant` (substantively discusses the specific NOAA-listed event – correct disaster, geography, and time window), `Borderline` (plausibly related but ambiguous, e.g. a brief wire item or a retrospective mention), or `Trash` (off-topic, wrong event, duplicate, paywall/landing page, dead link, or photo gallery with caption-only text). Only `Relevant` articles proceed to Step 2; `Borderline` and `Trash` articles are retained in the data for transparency but excluded from the index.

Step 2: Attribution coding. Each `Relevant` article is assigned to one of five categories describing how it frames the causal origin of the event (Table 1): (A) explicit anthropogenic attribution (+1.0), (B) hedged or contextual climate mention (+0.5), (C) no causal attribution (0.0), (D) explicit non-climate attribution to natural variability or other human causes (−1.0), and (E) mixed/contested framing (0.0, separately flagged as `Contested`). For category D, coders additionally record whether the non-climate attribution is to natural variability ("just weather") or to other human causes (e.g. land use, infrastructure), since both score −1 on the climate-vs-natural axis but imply different policy narratives.

Full article text retrieval. All coding—both the human calibration pass and the LLM full-sample pass—is performed on the complete article text rather than search-engine snippets. Full text was retrieved in two stages. First, archived copies of each article were obtained via archive.ph, a web archiving service that captures complete page content prior to paywall overlays. For articles unavailable via archive.ph, full text was retrieved manually via Factiva (Dow Jones). This two-stage retrieval procedure was adopted following a finding during calibration that critical attribution language can appear mid-article and is missed with snippet-only coding. Of the 1,694 candidate articles, full text was ultimately retrieved for 1,591 (93.9%); the remaining 103 are excluded from LLM coding and treated as missing in the Attribution Index, comprising 64 articles for which full text could not be recovered by either method (e.g. dead links, HTTP errors, or pages with no archived snapshot), 36 articles that resolved to photo/video-only gallery pages with no substantive article text, and

3 articles – all from `transcripts.cnn.com` – whose URL resolved to a generic chronological listing of segments rather than the specific transcript segment retrieved for that event.

Calibration and validation. Coding is performed by an LLM following the codebook. The LLM is Claude (model `claude-sonnet-4-6`; Anthropic, Inc.), accessed via the Anthropic Messages API, with the codebook supplied as a system prompt and the full article text as the user message. Validation follows a staged procedure that applies two different reliability bars: a low gate during calibration to catch codebook problems cheaply, and a higher target for the reliability statistics reported in the paper. First, the researcher hand-codes a stratified random calibration sample of 50 articles (stratified by disaster type and by pre-/post-2010 period) for both Relevance and Attribution category. Second, the LLM codes the same 50 articles from the codebook alone. Third, human and LLM codes are compared using unweighted Cohen's κ for the categorical Relevance (3-category) and Attribution (5-category) codes; if agreement is weak ($\kappa < 0.6$, "moderate" or worse on the Landis and Koch (1977) scale), the codebook wording and examples are revised and the comparison is repeated before scaling up. This 0.6 threshold is intentionally low—its purpose is to catch broken category definitions early, not to certify reliability.

The calibration gate was cleared on the initial pass. Unweighted κ for Relevance was 0.81 ("substantial agreement" on the Landis and Koch 1977 scale, exceeding the $\kappa \geq 0.7$ reporting target). For Attribution, both unweighted and linearly weighted κ were 1.00 (perfect agreement), with a mean absolute difference of 0.00 between human- and LLM-assigned numeric scores. The four residual Relevance disagreements were all boundary cases (`Trash` vs. `Borderline`) where the correct classification is genuinely ambiguous; the relevant decision rules have been noted in the codebook for the full-sample pass.

Once this gate was cleared, the LLM coded all remaining articles in the full sample (1,591 articles with retrievable full text; the 50 calibration-sample articles were not re-coded by the full pass and instead carry forward their calibration-stage LLM codes, for consistency of method across the full `Master` sheet).

Finally, an independent post-hoc audit sample is drawn from the fully coded data, excluded from the calibration sample, for a final human spot-check. Because attribution categories D and E are rare in the full corpus (4.2% and 0.8% of Relevant articles respectively), a simple random draw of the target 100 articles would likely contain too few instances of either to support a meaningful reliability statistic for them. The audit sample therefore deliberately oversamples these categories – including all 10 category-E articles and a random 20 of the 53 category-D articles – with the remaining 70 slots filled by a proportionally stratified random draw (by disaster type and pre-/post-2010 period) from the rest of the coded corpus, for a total of 100 articles. Because of this deliberate oversampling, the audit sample is not a simple random sample of the corpus; reliability statistics computed from it should be read as conditional on this design (informative about each category's reliability individually) rather than as a corpus-representative error rate, and this is noted explicitly when those statistics are reported. The audit sample is hand-coded blind to the LLM's codes and then compared to them. For Relevance we report unweighted Cohen's κ (target $\kappa \geq 0.7$); for Attribution, we report both unweighted κ and a weighted κ (linear weights) that reflects the ordinal structure of the underlying scores $\{-1, 0, 0.5, 1\}$, since unweighted κ treats an A-vs-D disagreement the same as a B-vs-C disagreement despite the very different implied score gap (target $\kappa_w \geq 0.7$, ideally ≥ 0.8); and we additionally report the mean absolute difference between human- and LLM-assigned numeric scores as a directly interpretable companion statistic. Should the audit-sample statistics fall short of these targets, this is disclosed and discussed as a limitation, since the boundaries between categories B/C and C/E are genuinely fuzzy even for human coders.

The audit sample has been completed. For Relevance (all 100 audited articles), raw agreement was 99% and unweighted Cohen's $\kappa = 0.96$, comfortably exceeding the $\kappa \geq 0.7$ target; the single disagreement was a Trash/Borderline boundary case. For Attribution, restricted to the 86 articles both the human coder and the LLM independently classified as Relevant (the subsample where both raters have a meaningful 5-category code), raw agreement was 95.3%, unweighted $\kappa = 0.93$, linearly weighted $\kappa_w = 0.96$, and quadratically weighted $\kappa_w = 0.98$ – all well above the $\kappa \geq 0.7$ target and the $\kappa_w \geq 0.8$ "ideal" threshold. The mean absolute difference between human- and LLM-assigned numeric scores was 0.023 on the $[-1, 1]$ scale. All four Attribution disagreements fell on adjacent categories on the causal-framing spectrum (two A/B, one B/C, one B/E) – exactly the kind of boundary fuzziness anticipated above – with zero disagreements spanning a larger gap (e.g. no A-vs-D confusions). Full counts and the complete confusion matrices are reproducible from `code/audit_reliability.py`.

4.2 Disaster-Level Attribution Index

For each disaster d with N_d articles coded `Relevant`, article-level scores are averaged to a disaster-level *Attribution Index*:

$$\text{AttributionIndex}_d = \frac{1}{N_d} \sum_{i=1}^{N_d} a_{i,d}, \qquad a_{i,d} \in \{-1,\ 0,\ 0.5,\ 1\} \tag{1}$$

where $a_{i,d}$ is the numeric score for article i covering disaster d from Table 1, with -1

indicating explicit non-climate (natural/random or other human-cause) attribution and +1 indicating explicit anthropogenic climate-change attribution. Category E (mixed/contested) scores 0, identically to category C (no attribution), but is additionally flagged so that a disaster-level *Contested share* – the fraction of N_d articles coded E – can be reported separately as a measure of framing disagreement, distinct from the simple absence of any causal claim.

For the 76 events with $N_d = 0$ (no relevant articles, including events that returned zero search results entirely), $\text{AttributionIndex}_d$ is left undefined (missing) rather than imputed, and these events are analyzed as a separate "no media attribution" group rather than dropped from the sample. All of the above quantities – N_d, $\text{AttributionIndex}_d$, and the Contested share – are computed via formulas in the `All Disasters` sheet of the project workbook (`attribution.xlsx`), which references the article-level codes in the `Master` sheet.

4.3 Specification

We relate the Attribution Index to disaster and outlet characteristics at two levels, deliberately. Disaster type, severity, geography, and time are properties of the *event* and do not vary across the articles covering it, so they are tested in a disaster-level cross-section. National/local status and outlet political lean, by contrast, vary *within* a disaster – the same wildfire is often covered by both a wire service and a local paper – so collapsing them to one disaster-level number would discard most of the identifying variation; these are instead tested in an article-level specification with standard errors clustered by disaster.

Disaster-level specification. For each disaster d with $N_d \geq 1$ (i.e. excluding the 104 events with no Relevant coverage):

$$\begin{aligned}\text{AttributionIndex}_d = \beta_0 + \boldsymbol{\delta}' \,\text{Type}_d + \beta_1 \ln(\text{Cost}_d) + \beta_2 \ln(\text{Deaths}_d + 1) + \beta_3 \ln(\text{Duration}_d) \\ + \beta_4 \,\text{CoastalShare}_d + \beta_5 \,\text{RedShare}_d + \beta_6 \,\text{Year}_d + \beta_7 \, N_d + \varepsilon_d \quad (2)\end{aligned}$$

where Type_d is a vector of disaster-type dummies (reference category: Severe Storm, the modal type); cost and deaths are log-transformed given their strong right skew (ln(Deaths+1) to accommodate zero-fatality events); CoastalShare_d and RedShare_d are, respectively, the share of d's affected states that border an ocean or the Gulf of Mexico and the share that voted Republican in the 2020 presidential election – shares rather than a single "primary state" because the `States` field lists affected states alphabetically, not by impact, so no reliable primary state can be recovered for multi-state events; and N_d (the number of Relevant articles) is included as a control for coverage volume. Standard errors are heteroskedasticity-robust (HC1).

Article-level specification. For each Relevant article i covering disaster d:

$$a_{i,d} = \gamma_0 + \gamma_1 \, \text{National}_i + \gamma_2 \, \text{OutletLean}_i + \boldsymbol{\delta}' \, \text{Type}_d + \gamma_3 \, \text{Year}_d + u_{i,d} \tag{3}$$

where National_i is the existing `National Newspaper` indicator and OutletLean_i is the outlet's Ad Fontes Bias score (continuous, -42 to $+42$, more positive = more right-leaning), joined by domain; standard errors are clustered by disaster d to account for non-independence among articles covering the same event. OutletLean_i is only defined for the 33 of 135 distinct domains in the sample with an individually-rated page on Ad Fontes/AllSides (covering 71% of Relevant articles, since coverage is concentrated in large outlets), so specifications including it are estimated on that subsample.

5 Results

All results below are estimated on the 209 disasters with $N_d \geq 1$ (or the relevant article-level subsample). The independent post-hoc audit described in Section 4 has been completed and clears the reliability targets ($\kappa = 0.96$ for Relevance, weighted $\kappa_w = 0.96$–0.98 for Attribution); see Section 6 for the full statistics.

5.1 Descriptive Patterns

Figure 1 plots the disaster-level Attribution Index against year, together with a fitted linear trend. The index is centered near zero (or negative) for most of the sample's first decade and rises steadily afterward: mean $\text{AttributionIndex}_d$ is -0.105 (SD 0.250) for disasters beginning before 2010, versus $+0.057$ (SD 0.230) from 2010 onward, a difference that is statistically significant (Welch $t = -3.63$, $p < 0.001$). A bivariate regression of the index on year gives a slope of 0.0129 per year ($p < 0.001$, robust SE), i.e. coverage has moved from net-skeptical/natural framing toward modestly net-anthropogenic framing over the sample period, though even at the end of the sample the average disaster's coverage remains much closer to "no causal claim" (category C) than to confident anthropogenic attribution.

Figure 2 shows the full cross-sectional distribution underlying that trend. The index is exactly 0.0 for 55.5% of covered disasters (consistent with category C, no causal attribution, being the modal article-level code), with 30.6% net-positive (mean tilt toward anthropogenic

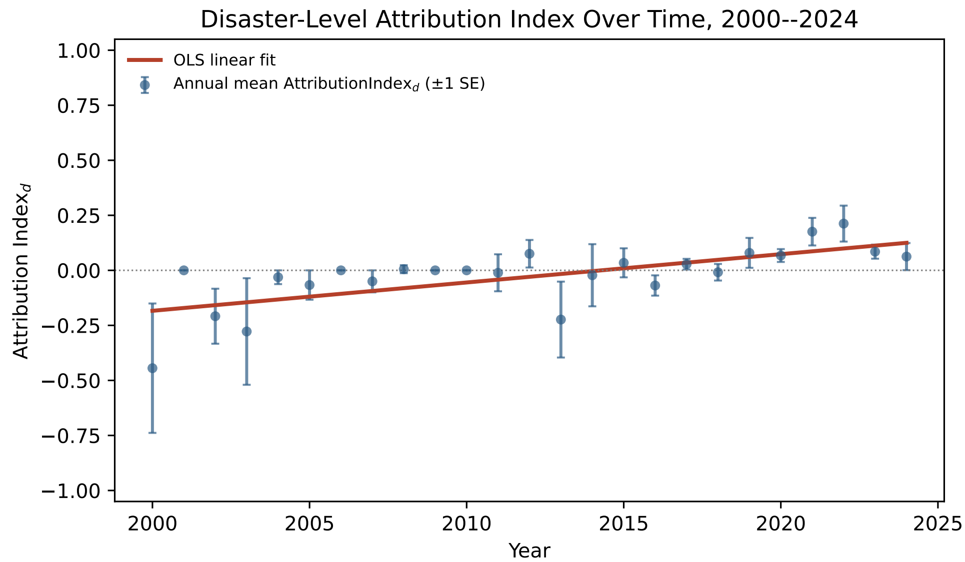

Figure 1: Attribution Index Over Time. Points are annual means across covered disasters ($\pm$1 SE); line is an OLS fit of AttributionIndex$_d$ on year (slope = 0.0129, $p < 0.001$).

framing) and 13.9% net-negative; most of the non-zero mass sits close to zero in both directions, and disasters with a strongly net-anthropogenic (AttributionIndex$_d$ near +1) or net-skeptical (near -1) coverage profile are rare. This reinforces the reading from Section 3.4: the year-over-year movement documented above is a shift in the modest tilt of an otherwise near-zero-centered distribution, not a shift between two opposed extremes.

5.2 Variation by Geography

Splitting disasters into "Coastal" (at least one affected state borders an ocean or the Gulf of Mexico) versus "Inland" shows no significant difference: mean Attribution Index is 0.023 (SD 0.235, $N = 179$) for Coastal disasters and 0.070 (SD 0.301, $N = 22$) for Inland disasters (Welch $t = -0.72$, $p = 0.48$). Classifying disasters by the 2020 presidential vote of their affected states (majority-rule across states, for disasters spanning more than one state) likewise shows essentially no difference: 0.023 (SD 0.247) for Majority-Blue disasters ($N = 71$) versus 0.023 (SD 0.239) for Majority-Red disasters ($N = 125$); the 5 Split/Tied disasters average higher (0.213) but the group is too small to draw any conclusion from. A regression of the index on the continuous CoastalShare$_d$ and RedShare$_d$ measures (which use the fraction of a multi-state disaster's states that are coastal/Republican, rather than collapsing to a single category) confirms both are statistically indistinguishable from zero (Table 5, columns 2–3). Eight disasters with `States = NATIONAL` or limited to Puerto Rico (which has no 2020 presidential vote) are excluded from this section and from any specification including

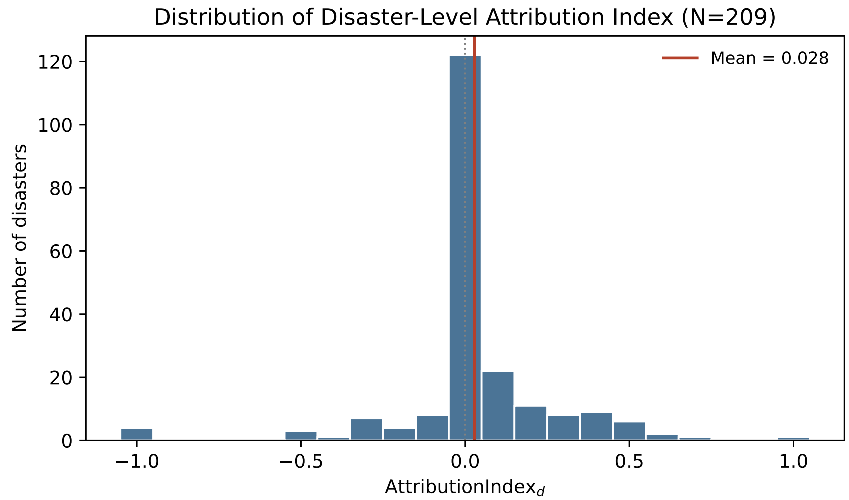

Figure 2: Distribution of the Disaster-Level Attribution Index ($N = 209$ covered disasters). Vertical line at the mean (0.028); dotted line at zero.

geography.

5.3 Variation by Disaster Type and Severity

Mean Attribution Index varies little across NOAA disaster types – from 0.000 (Freeze, $N = 2$) to 0.074 (Winter Storm, $N = 12$) – and a one-way ANOVA finds no significant difference across the seven types ($F = 0.32$, $p = 0.93$; full per-type means in Table 3, full type-dummy regression coefficients in Appendix Table 7). Severity, measured as log-CPI-adjusted cost and log(deaths+1), is similarly uninformative on its own (correlation with the index of 0.09 and 0.07 respectively, neither coefficient significant at conventional levels in Table 5); event duration (log days) is also not a significant predictor. In short, none of the disaster's own physical characteristics – what kind of disaster it was, how costly, how deadly, or how long it lasted – predicts how its coverage frames causation, at least in this disaster-level cross-section.

5.4 Variation by Outlet Characteristics

Unlike disaster type, severity, and geography, outlet characteristics show up clearly – but only once a confound is accounted for. In a simple bivariate comparison, articles from national outlets score modestly higher than local outlets (0.120 vs. 0.077 on the $[-1, 1]$ scale; clustered t-test $p = 0.079$), but this gap is not robust to controlling for disaster type and year: in the multivariate article-level specification (Table 4, column 2), the National coefficient falls to

Table 3: Mean Attribution Index by Disaster Type

Disaster Type	N	Mean	Median	SD
Severe Storm	89	0.005	0.000	0.236
Tropical Cyclone	45	0.055	0.022	0.113
Flooding	25	0.028	0.000	0.175
Drought	18	0.026	0.017	0.476
Wildfire	18	0.048	0.021	0.315
Winter Storm	12	0.074	0.000	0.148
Freeze	2	0.000	0.000	0.000

Notes: One-way ANOVA across the seven types: $F = 0.32$, $p = 0.93$.

0.031 and is no longer statistically distinguishable from zero ($p = 0.46$). National outlets in this sample skew toward later years and toward disaster types that independently trend toward more anthropogenic framing, and that composition effect appears to be doing the work in the bivariate comparison.

Outlet political lean tells a different story. Using the Ad Fontes Bias score (continuous, more positive = more right-leaning) for the 71% of Relevant articles from an individually-rated domain, a one-point increase in Bias score is associated with a 0.011-point *decrease* in Attribution Index ($p = 0.013$, clustered SE), controlling for disaster type, year, and national/local status (Table 4, column 2) – i.e. articles from more right-leaning outlets are systematically less likely to frame disasters as anthropogenically caused, even after accounting for which disasters and which years they're covering. This holds up in the bivariate specification as well (-0.013, $p = 0.002$) and is robust to using the AllSides score instead of Ad Fontes (-0.033, $p = 0.003$; Section 6). Figure 3 plots the underlying relationship: a jittered scatter of the four discrete article-level scores against outlet Bias score, with the OLS fit and binned means overlaid – the binned means track the linear fit closely, suggesting the relationship is not an artifact of a nonlinear pattern being forced into a straight line. (Section 6 also shows this effect is concentrated in national-outlet coverage and does not hold among local-only articles.)

5.5 Main Regression Results

Table 5 reports the disaster-level specification (equation 2) in three nested columns: (1) disaster type and severity only; (2) adding geography and duration; (3) the full specification, adding year and coverage volume (N_d). Disaster type, cost, deaths, duration, and geography are not significant in any column. Year is strongly significant once added ($\beta = 0.0128$,

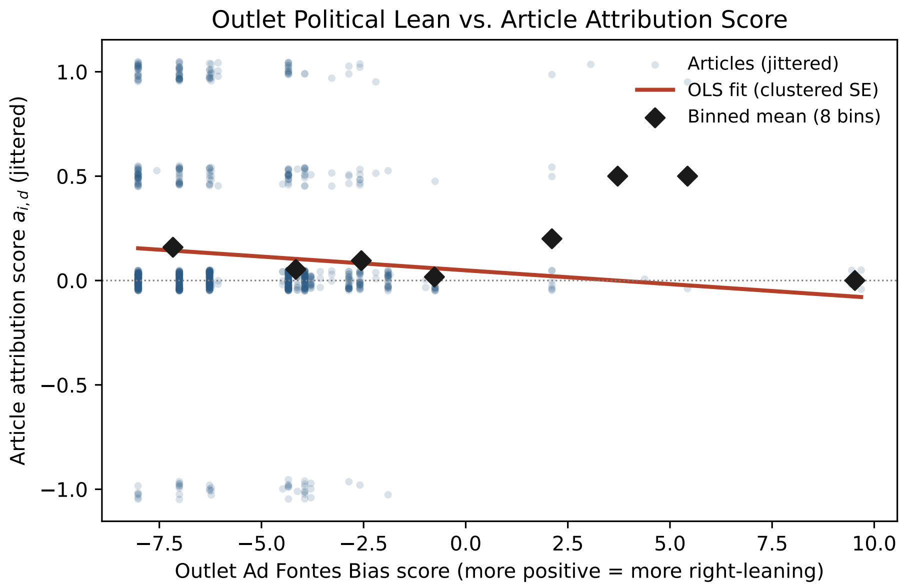

Figure 3: Outlet Political Lean vs. Article Attribution Score ($N = 889$ articles from individually-rated domains). Points are jittered on the y-axis for visibility, since the underlying score only takes four values $(-1, 0, 0.5, 1)$; diamonds are means within 8 equal-width bins of Ad Fontes Bias score.

$p < 0.001$), consistent with Figure 1, and coverage volume N_d is also positive and significant ($\beta = 0.0089$, $p = 0.036$) – disasters that draw more press coverage skew slightly more anthropogenic on average, though the effect is small and we would caution against a causal reading (more coverage could itself be a consequence of a disaster having a salient climate angle, rather than a cause of one). The full specification explains 20.1% of disaster-level variance in the Attribution Index, almost entirely attributable to the year and N_d terms – the disaster's own type, cost, death toll, and geography contribute almost nothing on top of those.

6 Robustness

Alternative outlet-lean measure. The outlet political-lean result in Section 5.4 uses Ad Fontes' numeric Bias score because every one of the 33 individually-rated domains in the sample has a clean numeric value there, whereas a handful of AllSides ratings returned only a category label with no exact meter value (mapped, for this check, to a fixed scale: Left $= -5$, Lean Left $= -2.5$, Center $= 0$, Lean Right $= +2.5$, Right $= +5$). Re-estimating the bivariate article-level specification with the AllSides score in place of Ad Fontes gives the same sign and a comparable magnitude (-0.033, $p = 0.003$, clustered SE, $N = 930$), so the

Table 4: Outlet Characteristics: Article-Level Results

	(1)	(2)
	Dep. var.: $a_{i,d}$ (article attribution score)	
National outlet	0.0308	0.0308
	(0.0418)	(0.0418)
Ad Fontes Bias score		-0.0112^{**}
		(0.0045)
Year	0.0169^{***}	0.0177^{***}
	(0.0024)	(0.0027)
Disaster type dummies	Yes	Yes
Clustered SE (by disaster)	Yes	Yes
N (articles)	1,252	889
Clusters (disasters)	209	194
R^2	0.112	0.124

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Standard errors in parentheses, clustered by disaster. Column (1) is estimated on all 1,252 Relevant articles; column (2) is restricted to the 889 articles (71%) from one of the 33 individually-rated domains, since Ad Fontes lean is undefined otherwise. Full disaster-type dummy coefficients are in Appendix Table 8.

political-lean finding is not an artifact of which rating service is used.

Disaster fixed effects. The article-level outlet-lean specification in Table 4 controls for disaster type and year but only *clusters* standard errors by disaster, leaving open the possibility that some other disaster-level confound drives the result. Re-estimating with disaster fixed effects – so that National_i and OutletLean_i are identified purely from variation *within* the same disaster (e.g. the AP vs. a local paper covering the same wildfire), implemented via within-disaster demeaning on the 144 disasters with ≥ 2 lean-rated articles – leaves the lean coefficient intact (-0.012, $p = 0.046$) and the national coefficient modestly larger and now marginally significant (0.075, $p = 0.099$). This is the most demanding test available in this sample, and the lean result survives it.

Alternative attribution measures. Three checks probe whether the year trend (Section 5) and the outlet-lean effect (Section 5.4) depend on the specific $\{-1, 0, 0.5, 1\}$ scoring of the Attribution Index. First, collapsing to a binary "any climate mention" (A/B) vs. "none" (C/D/E) measure leaves both findings significant (year trend $\beta = 0.0100$, $p < 0.001$; outlet lean $\beta = -0.0177$, $p < 0.001$). Second, recoding category-D articles attributed to *other human causes* (rather than natural variability) from -1 to 0 – since that framing is

Table 5: Determinants of the Attribution Index (Disaster Level)

	(1)	(2)	(3)
	Dep. var.: AttributionIndex$_d$		
ln(Cost)	0.0226	0.0254	0.0120
	(0.0188)	(0.0203)	(0.0175)
ln(Deaths + 1)	−0.0061	−0.0057	0.0045
	(0.0182)	(0.0191)	(0.0166)
ln(Duration)		−0.0203	0.0012
		(0.0214)	(0.0213)
CoastalShare$_d$		−0.0191	−0.0214
		(0.0664)	(0.0628)
RedShare$_d$		0.0338	0.0164
		(0.0553)	(0.0549)
Year			0.0128***
			(0.0036)
N_d (coverage volume)			0.0089**
			(0.0042)
Disaster type dummies	Yes	Yes	Yes
N (disasters)	201	201	201
R^2	0.015	0.019	0.201

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Heteroskedasticity-robust (HC1) standard errors in parentheses. Sample restricted to the 201 (of 209) covered disasters with classifiable geography (excludes 8 disasters coded `States = NATIONAL` or Puerto Rico only). Reference disaster type is Severe Storm; full type-dummy coefficients are in Appendix Table 7.

not really climate-skeptical, just non-climate – leaves both findings essentially unchanged (year trend $\beta = 0.0102$, $p < 0.001$; outlet lean $\beta = -0.0135$, $p < 0.001$). Third, dropping the 10 category-E (contested) articles entirely, rather than scoring them 0, again leaves both findings unchanged (year trend $\beta = 0.0130$, $p < 0.001$; outlet lean $\beta = -0.0135$, $p = 0.001$).

Sample restrictions. Restricting the disaster-level sample to the 138 disasters with $N_d \geq 3$ (so the index is not a single article's score) leaves the year trend significant ($\beta = 0.0112$, $p < 0.001$). Restricting the outlet-lean test to post-2010 articles only – since the pre-2010 era contributes just 37 of 209 covered disasters and could otherwise be driving the result – likewise leaves it significant ($\beta = -0.0149$, $p = 0.001$, $N = 779$).

Placebo checks. If outlet political lean is capturing climate-framing specifically, it should not also predict things that have nothing to do with framing. Two such checks are run. First, lean against article word count: this is *not* a clean null ($\beta = -18.7$ words per Bias-score point, $p = 0.075$) – more right-leaning outlets in this sample write modestly shorter articles, so part of the lean-attribution relationship plausibly reflects a generic length/depth difference across outlets rather than climate-framing decisions specifically, a caveat we flag rather than paper over. Second, lean against the covered disaster's (NOAA-reported, not outlet-chosen) log-cost and log-deaths: both are clean nulls ($p = 0.685$ and $p = 0.300$ respectively), meaning more right- or left-leaning outlets are not systematically selecting into covering more or less severe disasters – the lean effect is not a selection-into-coverage artifact, even if it may partly reflect a style/depth difference per the word-count check.

National-only vs. local-only subsamples. A different question from Section 5.4 (does national status itself predict the score) is whether the two headline effects are *homogeneous* across national and local coverage, or concentrated in one segment. Re-estimating each headline regression separately by outlet type: the year trend holds in both ($\beta = 0.0174$, $p < 0.001$ among national-outlet articles; $\beta = 0.0107$, $p = 0.007$ among local-outlet articles). The outlet-lean effect does not generalize the same way – it is strongly significant among national-outlet articles ($\beta = -0.0308$, $p = 0.001$, $N = 687$) but statistically indistinguishable from zero among local-outlet articles ($\beta = -0.0025$, $p = 0.69$, $N = 202$). This qualifies the political-lean finding meaningfully: it is robust to every measurement and specification choice tried above, including the demanding disaster- fixed-effects test, but appears to be concentrated in (or driven by) national/wire-service coverage rather than holding uniformly across local outlets as well – though the local subsample is also a third the size of the national one, so this could partly reflect lower power rather than a true zero effect locally. We report this as an open qualification rather than resolve it, since distinguishing the two would require a larger sample of lean-rated local outlets than is currently available.

Ordered logit/probit (functional-form check). Every result above treats $a_{i,d} \in \{-1, 0, 0.5, 1\}$ as cardinal via OLS – a real assumption, since the gap between "no claim" (C) and "hedged mention" (B) need not be the same size as the gap between B and "explicit attribution" (A). Dropping this assumption entirely and re-estimating both headline relationships as an ordered outcome (collapsing C and E to one level, since both score 0: $D < \{C, E\} < B < A$), via both ordered logit and ordered probit, leaves both findings significant under all four combinations: the year trend (logit $\beta = 0.104$, $p < 0.001$; probit $\beta = 0.055$, $p < 0.001$) and the outlet-lean effect (logit $\beta = -0.085$, $p = 0.004$; probit

$\beta = -0.043$, $p = 0.006$). Neither result is an artifact of the cardinal-scaling assumption.[2]

Year $\times$ outlet-lean interaction. Has the political-lean gap in framing widened or narrowed over the sample period? Adding an $\text{OutletLean}_i \times \text{Year}_d$ interaction term to the article-level model, the interaction is not statistically significant either alone ($\beta = -0.00028$, $p = 0.61$) or with the full set of controls ($\beta = -0.00030$, $p = 0.66$). The point estimate implies a slightly more negative (i.e. modestly larger-magnitude) lean effect by the end of the sample than the beginning, but this is not distinguishable from no change given the standard error – we find no evidence the ideological gap in attribution framing has systematically widened or narrowed over 2000–2024.

Table 6 summarizes all checks. With the exception of the word-count placebo and the local-only outlet-lean subsample, every alternative specification and sample restriction leaves both headline results intact.

Audit results. All results in Section 5 are estimated from the full-sample LLM coding pass. The independent post-hoc audit sample (100 articles, described in Section 4) has now been hand-coded and compared against the LLM's codes. Relevance agreement was 99% raw / $\kappa = 0.96$ (target $\kappa \geq 0.7$); Attribution agreement, on the 86 articles both raters independently classified as Relevant, was 95.3% raw, unweighted $\kappa = 0.93$, linearly weighted $\kappa_w = 0.96$, and quadratically weighted $\kappa_w = 0.98$ (targets $\kappa \geq 0.7$, ideally $\kappa_w \geq 0.8$), with a mean absolute score difference of 0.023 on the $[-1, 1]$ scale. All four Attribution disagreements were between adjacent categories (two A/B, one B/C, one B/E); none spanned a larger gap. These confirm, on an independent sample, the high reliability already suggested by the calibration-sample agreement ($\kappa = 0.81$ for Relevance, $\kappa = 1.00$ for Attribution) – the results in Section 5 can be read as confirmed rather than provisional.

7 Discussion

The Year Trend and the 2010 Structural Break

The most robust finding in this paper is a steady upward trend in anthropogenic attribution framing: covered disasters average an Attribution Index of -0.105 before 2010 and $+0.057$ from 2010 onward, a shift of more than 0.16 index points that persists under every robustness check we apply (Table 6). This timing aligns closely with the structural break identified by

[2]The ordered-model standard errors here are not clustered by disaster (the `statsmodels` ordered-model implementation does not support cluster-robust SE), so these p-values are an approximate check on sign and significance rather than directly comparable to the clustered-SE results elsewhere in this section.

Chinn et al. (2020), who find using unsupervised text scaling (Wordfish) that partisan *polarization* in U.S. climate news language was relatively flat through the 2000s before undergoing a discrete jump in 2010–2011. Our result complements theirs: not only did the language of climate coverage polarize around 2010, but the specific practice of linking extreme weather events to anthropogenic causes — which had been net-negative through the early 2000s — shifted decisively toward net-positive after that point.

What drove the post-2010 shift? This paper identifies the *when* more cleanly than the *why*, and several mechanisms are plausible. First, the scientific literature on extreme-event attribution expanded substantially after roughly 2011, when formal attribution science (asking whether anthropogenic forcing changed the probability of a specific event) began receiving mainstream scientific attention; growing scientific output on individual events plausibly gave journalists more attributable claims to report. Second, Chinn et al. (2020) document that political actors outnumber scientists as quoted sources in U.S. climate news from 2006 onward, suggesting that supply-side editorial choices — which voices are treated as authoritative — shift the content of coverage independently of the underlying science. Third, the period after 2009–2010 saw a series of individually prominent extreme events (major flooding, record droughts, Hurricane Sandy) that attracted sustained national attribution commentary, potentially establishing a template for how subsequent events were covered. Disentangling these mechanisms would require event-level attribution-science publication dates matched to coverage timing, which is beyond the scope of the present paper but is a natural extension.

Outlet Political Lean: Supply, Demand, and the National–Local Asymmetry

The second main finding is that more right-leaning outlets (by Ad Fontes Bias score) produce systematically lower attribution scores, with an estimated gradient of -0.011 per Bias-score point ($p = 0.013$, clustered SE), robust to disaster fixed effects, both lean measures, all attribution-measure variants, and the ordered-logit functional form. Two broad explanations are consistent with this pattern: a supply-side story in which editorial decisions reflect the ideological preferences of editors, owners, or reporters; and a demand-side story in which outlets cater to the beliefs of their readers.

Gentzkow and Shapiro (2010) provide relevant evidence on this question in the context of general media slant: using the frequency of partisan phrases in congressional speech to construct outlet slant indices, they find that approximately 20% of cross-outlet variation in slant is explained by reader demographic demand, while owner identity explains substantially less. If the same demand-side logic applies to attribution framing specifically, much of the lean gradient we document may reflect outlets supplying the causal narratives their audiences expect: right-leaning outlets serve audiences who, consistent with Carmichael et al. (2017)'s evidence on partisan media and climate concern, are less receptive to anthropogenic attribution framing.

The national–local asymmetry qualifies this picture importantly. The lean gradient is strongly significant among national-outlet articles ($\hat{\beta} = -0.031$, $p = 0.001$, $N = 687$) but is statistically indistinguishable from zero among local-outlet articles ($\hat{\beta} = -0.003$, $p = 0.69$, $N = 202$). The local subsample is about one-third the size of the national one, so part of this difference may reflect lower power rather than a true zero effect among local outlets. Nonetheless, it raises the possibility that local papers — which serve geographically specific audiences who may have directly experienced an event — behave differently from national and wire outlets when deciding how to frame causation. Ellger et al. (2024) show, in the context of German newspaper exits, that local papers play a distinct informational role that is not substitutable by national coverage: when a local paper exits, consumers shift to national tabloids that are more partisan and less locally focused, and political polarization rises as a consequence. A symmetric reading of their result applied here would suggest that local papers may moderate partisan attribution framing relative to national outlets, and that the ongoing contraction of local print journalism could shift the aggregate information environment further toward the nationally partisan lean gradient we document.

The non-significance of the year $\times$ lean interaction ($p = 0.61$ in the simple model, $p = 0.66$ with full controls) adds a further nuance: the ideological gap in attribution framing has not significantly widened or narrowed over the sample period. Whatever mechanism generates the lean gradient appears to have been stable across the 2000–2024 window, even as the overall level of attribution framing rose.

Null Results on Disaster Characteristics

A striking feature of the results is the near-complete absence of predictive power from the disaster's own physical characteristics. Disaster type, CPI-adjusted cost, death toll, and duration are all statistically indistinguishable from zero in the disaster-level specification (Table 5), and a one-way ANOVA across the seven NOAA event types confirms no significant variation ($F = 0.32$, $p = 0.93$; Table 3). This null result on disaster type is consistent with Boudet et al. (2020), who find that even events with a high scientific attribution confidence score (such as extreme heat) do not reliably generate more attribution discussion at the local level than events with lower attribution confidence. In both their study and ours, the content of coverage appears more responsive to contextual factors outside the event itself — timing, outlet identity, and the broader information environment — than to the physical or scientific characteristics of the disaster being covered.

This finding also has a methodological implication. If disaster type and severity do not predict attribution framing in the cross-section, it is unlikely that the composition of NOAA events across years (e.g., more tropical cyclones in some years than others) is what drives the year trend. The year trend appears to be a genuine temporal shift in how coverage is framed, not a composition artifact.

Coverage Volume

Coverage volume (N_d, the number of Relevant articles per disaster) enters the disaster-level regression positively and significantly ($\hat{\beta} = 0.009$, $p = 0.036$): disasters that attract more press coverage skew modestly more anthropogenic on average. This association admits two interpretations that are difficult to distinguish in observational data. On the first, a high climate-angle salience increases both the volume of coverage and the probability of attribution framing — the same editorial judgment that generates more articles also generates more attribution language, so the two are joint outcomes of latent salience. On the second, with more articles covering a given event there is a higher probability that at least one of them introduces attribution framing that then pulls the disaster- level average above zero, a purely mechanical relationship. Given this ambiguity, we control for N_d throughout the analysis but do not interpret the coefficient causally.

Implications for Risk Perception and Policy

The pattern we document — near-zero average Attribution Index, a rising trend post-2010, and a partisan lean gradient concentrated in national coverage — has several implications for the larger research programs on disaster risk perception and preparedness.

Gallagher (2014) shows that media coverage of flood events materially shifts insurance take-up, both in directly affected areas (9% spike) and in non-affected communities in the same television media market (one-third of the direct effect). Critically, he finds that this salience channel operates at the same magnitude for high- and low-cost floods — the dollar size of the event does not drive the risk-learning response, the coverage does. If attribution framing further shapes the mechanism through which risk is perceived (is this disaster a precedent for future events, or a one-off?), then the near-zero and historically non-anthropogenic framing we document for most of the sample period may have dampened the longer-run risk signaling that coverage could otherwise have provided.

Magontier (2020) finds a related result on the policy side: a one-standard-deviation increase in storm media coverage is associated with 54% more FEMA hazard mitigation projects in the affected area, consistent with a risk-signaling channel from coverage to government preparedness investment. Whether the *content* of coverage — natural-variability framing versus anthropogenic framing — modulates the direction or magnitude of this preparedness response is an important open question our paper motivates but cannot answer directly, since it would require the mitigation-project data to be crossed with article-level attribution codes at the event level.

At the individual level, Egan and Mullin (2012) document that personal exposure to anomalously high local temperatures increases the probability that a respondent believes there is solid evidence of warming, with the effect largest among those who are less educated and who have weak attachments to political parties. The fact that this effect decays within 12 days suggests that individual-level risk updating from direct experience is short-lived, which raises the relative importance of media framing as a more durable information source. If right-leaning coverage is systematically less likely to frame extreme events as anthropogenically caused — as we find — then the populations that rely most on national right-leaning outlets for information about climate may receive a systematically different signal about the causes of the disasters they experience.

Limitations

Several limitations should inform how the findings are read. First, the Attribution Index is an average over a small and potentially unrepresentative sample of articles: 33.2% of NOAA events have no Relevant coverage at all, and the articles we do have are concentrated in large national and wire-service outlets. Whether the pattern we document in covered disasters generalizes to the median NOAA event is uncertain.

Second, the outlet-lean gradient is identified from the 33 domains with Ad Fontes or AllSides ratings, covering 71% of Relevant articles but concentrated in the largest outlets. The local-outlet subsample with lean ratings ($N = 202$ articles) is small enough that the null effect there could reflect limited power rather than a true zero.

Third, the LLM coding procedure, while validated by the post-hoc audit ($\kappa = 0.93$ unweighted, $\kappa_w = 0.96$–0.98 weighted), relies on a single model and a single codebook. The four Attribution disagreements in the audit were all on adjacent-category boundaries (A/B, B/C, B/E), which is the kind of fuzziness inherent in any ordinal text classification scheme, but it implies that the index is somewhat less reliable near the B/C boundary than at the extremes.

Fourth, the outlet-lean finding, while robust across all specification checks, is correlational. The most we can say is that more right-leaning outlets produce less anthropogenic attribution framing within the same disaster and year; we cannot rule out that a common cause drives both outlet lean and coverage decisions without controlling for it. The word-count placebo ($p = 0.075$) raises the additional possibility that some of the lean gradient reflects a generic length-and-depth difference across outlet types rather than a climate-framing decision specifically.

8 Conclusion

This paper constructs the first systematic, large-scale measure of how U.S. newspapers frame the causal origins of billion-dollar climate disasters. Using an LLM-based coding pipeline validated against an independent human audit ($\kappa = 0.96$ for Relevance, $\kappa_w = 0.96$–0.98 for Attribution), we assign each of 1,252 Relevant articles — collected across all 313 NOAA billion-dollar disaster events from 2000 to 2024 — to one of five attribution categories ranging from explicit anthropogenic framing (+1) to explicit non-climate framing (-1), and aggregate these to a disaster-level Attribution Index.

Three findings stand out. First, the dominant mode of coverage is silence on causation: 74.8% of Relevant articles contain no causal attribution at all (category C), and the mean disaster-level Attribution Index is only 0.028 — statistically above zero but economically near the neutral midpoint. The picture that emerges is not one of a media landscape polarized between climate advocates and climate skeptics, but of one in which the large majority of disaster coverage simply does not engage the question of cause.

Second, the level of attribution framing has risen steadily over the sample period, at a rate of approximately 0.013 index points per year ($p < 0.001$), with a particularly sharp shift centered on 2010. This timing coincides with the discrete structural break in partisan climate discourse identified by Chinn et al. (2020), and is robust to every functional-form and sample-restriction check we apply. By the end of the sample, average coverage has moved from net-negative (implying mild non-climate framing in aggregate) to modestly net-positive, though the dominant category remains pure event reporting.

Third, outlet political lean is a significant predictor of attribution framing. Articles from more right-leaning outlets (by Ad Fontes Bias score) receive lower Attribution Index values ($\hat{\beta} = -0.011$ per Bias-score point, $p = 0.013$, clustered SE), a result that survives disaster fixed effects, both lean-rating services, all attribution-measure variants, and an ordered-logit functional-form check. The effect is concentrated in national and wire-service coverage and is not statistically detectable in the local-outlet subsample, though the local subsample is small enough that power rather than a true zero cannot be ruled out. Notably, the political-lean gap has not widened or narrowed over the sample period: the year $\times$ lean interaction is not significant, suggesting that the ideological gradient in attribution framing has been stable even as the overall level has risen.

In contrast, none of the disaster's own physical characteristics — type, CPI-adjusted cost, death toll, duration, coastal geography, or political lean of affected states — predict how its coverage frames causation. This null result on disaster characteristics implies that the temporal and outlet-level patterns we identify are not composition artifacts: it is not that the mix of events changed after 2010 in a way that mechanically increased attribution framing, nor that right-leaning outlets happen to cover less climate-salient disasters.

Contributions. The paper makes three contributions. Methodologically, it demonstrates that a carefully validated LLM coding pipeline — following the three-step text-as-data framework of Gentzkow et al. (2019) — can reliably classify fine-grained causal attribution in news text at a scale ($N > 1{,}000$ articles) that would be prohibitively costly to hand-code, with inter-rater reliability exceeding the targets set in advance. Substantively, it produces the first panel of attribution framing across the full universe of NOAA billion-dollar disasters, enabling the year-trend and outlet-lean analyses that are the paper's headline results. Empirically, it connects the framing literature to the economics literature on disaster salience: given that Gallagher (2014) shows media coverage of flood events materially shifts risk-protective behavior even in non-affected communities in the same media market, the attribution content of that coverage is a potentially important and previously unmeasured input into how households and governments update their beliefs and preparedness decisions.

Limitations and future directions. Several limitations point toward natural extensions. The sample is constrained to events meeting NOAA's billion-dollar threshold (in real 2024 dollars), excluding the much larger universe of smaller disasters; whether attribution framing patterns differ for sub-threshold events is unknown. The outlet universe, while broad (899 verified domains), is biased toward outlets large enough to be indexed by Google's search engine; smaller community papers are underrepresented, and the lean- rated subsample is further concentrated in the largest national outlets. Extending the analysis to a more representative sample of local coverage — particularly as the local-newspaper industry continues to contract (Ellger et al., 2024) — would clarify whether the national–local asymmetry in lean effects we document is a durable structural feature or a power artifact.

Future work could also link the Attribution Index directly to downstream outcomes. Magontier (2020) shows that storm media coverage volume predicts FEMA mitigation investment; a natural next step is to ask whether the *content* of coverage — anthropogenic versus natural-variability framing — independently predicts preparedness, insurance take-up, or political responses. On the methodology side, an obvious extension is to apply the same pipeline to other countries' media markets, where the mapping between outlet lean and attribution framing may differ from the U.S. pattern given different partisan polarization dynamics and media market structures.

Finally, the year-trend finding raises a question the paper cannot answer: how much of the post-2010 rise in attribution framing reflects the growth of formal event-attribution science (which gave journalists attributable claims to cite), how much reflects shifting elite cues from politicians and advocacy organizations, and how much reflects demand-side changes in audience interest. Answering this would require matching attribution-science publication dates, politician attribution statements, and audience engagement metrics to article-level coverage timing — a demanding but tractable research design that would substantially clarify the mechanisms behind the trend documented here.

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A Data Appendix

A.1 Data and Code Availability

All data and code underlying this paper are available in the project's GitHub repository, github.com/Hariaksha/climate-disaster. The central data file – the full `Master` sheet of article-level codes, the `All Disasters` disaster-level table, the `Calibration` and `Audit Sample` sheets, and the `Domain Ratings` outlet-lean table – is `attribution.xlsx`. The Stage 3 regression analysis reported in Sections 5–6 is fully reproducible from `code/Stage3_Regressions.ipynb`, and the attribution-coding codebook is `paper/attribution_codebook.md`. Note that `attribution.xlsx` includes full retrieved article text (`Master` column `Full Article Text`), which is copyrighted newspaper content; this is shared here for replication purposes, but re-publication of that column specifically beyond replication use is not authorized by the original publishers.

A.2 Disaster Event List

The full list of 313 NOAA events, with assigned states/regions and PSE engine coverage, is the `All Disasters` sheet of `attribution.xlsx` (not reproduced in full here; see Section 3 for summary statistics).

A.3 Newspaper Source List

The 19 Programmable Search Engines and their domain coverage are documented in `data/Access World News Database.xlsx`; the resulting outlet-level political-lean ratings (AllSides, Ad Fontes) used in Section 5.4 are the `Domain Ratings` sheet of `attribution.xlsx`.

A.4 Additional Tables and Figures

The full Stage 3 analysis – dataset construction, all regressions and cross-tabs reported in Section 5, plus the robustness checks and placebo tests in Section 6 – is reproducible from `code/Stage3_Regressions.ipynb` (see Section A.1 for the full data/code availability statement).

Table 6: Robustness Check Summary

Check	Estimate	p-value
Year trend		
Baseline	0.0129/yr	< 0.001
Binary attribution measure	0.0100	< 0.001
D-subtype recoded	0.0102	< 0.001
Contested (E) excluded	0.0130	< 0.001
$N_d \geq 3$ only	0.0112	< 0.001
National-outlet articles only	0.0174	< 0.001
Local-outlet articles only	0.0107	0.007
Ordered logit	0.1044	< 0.001
Ordered probit	0.0546	< 0.001
Outlet political lean (Ad Fontes Bias score)		
Baseline (clustered SE)	−0.0132	0.002
Disaster fixed effects	−0.0115	0.046
Binary attribution measure	−0.0177	< 0.001
D-subtype recoded	−0.0135	0.001
Contested (E) excluded	−0.0135	0.001
Post-2010 only	−0.0149	0.001
National-outlet articles only	−0.0308	0.001
Local-outlet articles only	−0.0025	0.689
Ordered logit	−0.0850	0.004
Ordered probit	−0.0434	0.006
× Year interaction	−0.00028	0.611
Placebo checks (outlet lean predicting non-framing outcomes)		
Word count	−18.69	0.075
log(Cost) of covered disaster	−0.012	0.685
log(Deaths+1) of covered disaster	−0.034	0.300

Notes: Year-trend checks are disaster-level OLS, robust (HC1) SE, except the ordered logit/probit rows, which are article-level (ordinal attribution category regressed on year, non-clustered MLE SE). Outlet- lean and placebo checks are article-level OLS, SE clustered by disaster (or within-disaster-demeaned with no intercept, for the fixed-effects row), except the ordered logit/probit and interaction rows (non-clustered MLE SE and clustered OLS, respectively – see notebook for exact specifications). Full detail and code for every row in `code/Stage3_Regressions.ipynb`, Section K.

Table 7: Full Disaster-Type Coefficients, Disaster-Level Model (Column 3 of Table 5)

Disaster Type (ref. = Severe Storm)	Coef.	SE
Drought	0.0050	(0.1260)
Flooding	0.0195	(0.0488)
Freeze	0.1119	(0.0667)
Tropical Cyclone	0.0106	(0.0556)
Wildfire	0.0407	(0.0903)
Winter Storm	0.0790	(0.0695)

Notes: Robust (HC1) SE. None of the six type dummies is individually significant at $p < 0.10$. Companion coefficients (cost, deaths, duration, geography, year, N_d) are in Table 5, column 3.

Table 8: Full Disaster-Type Coefficients, Article-Level Model (Column 2 of Table 4)

Disaster Type (ref. = Severe Storm)	Coef.	SE
Drought	0.3412***	(0.1054)
Flooding	0.1020**	(0.0518)
Freeze	0.1442**	(0.0708)
Tropical Cyclone	0.1203**	(0.0478)
Wildfire	0.2411***	(0.0776)
Winter Storm	0.0663	(0.0481)

Notes: $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. SE clustered by disaster. Unlike the disaster-level model, disaster type is strongly significant here: at the article level, with disaster-level confounds (cost, geography, etc.) absorbed differently and N roughly 4–6× larger, droughts, wildfires, tropical cyclones, floods, and freezes all score significantly higher than severe storms. Companion coefficients (National, Ad Fontes Bias, Year) are in Table 4, column 2. The contrast with Table 7 (where type is insignificant) reflects the different unit of analysis and sample restriction (889 articles from rated domains vs. 201 disasters) and should not be over-read as a contradiction.